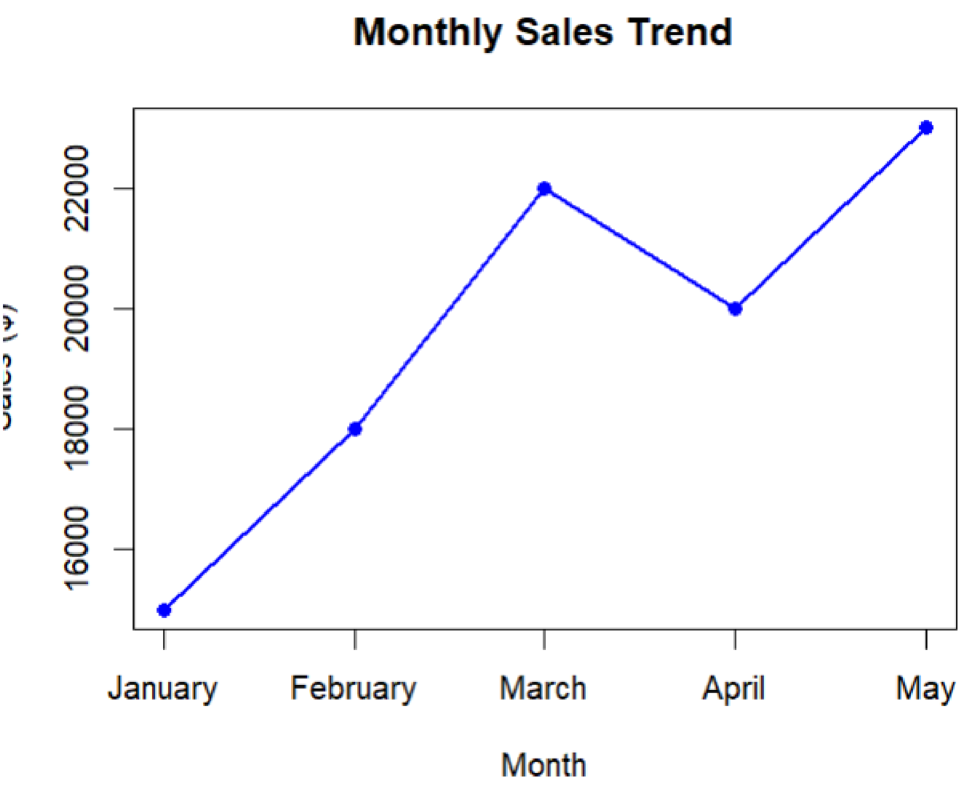
Exp1: 1: Monthly Sales Data

a) R code:

```
# Dataset
sales <- data.frame(
  Month = c("January", "February", "March", "April", "May"),
  Sales = c(15000, 18000, 22000, 20000, 23000)
)

# Line Chart
plot(sales$Sales,
     type = "o",
     xaxt = "n",
     xlab = "Month",
     ylab = "Sales ($)",
     main = "Monthly Sales Trend",
     col = "blue",
     lwd = 2,
     pch = 16)

axis(1,
     at = 1:nrow(sales),
     labels = sales$Month)
```

output:

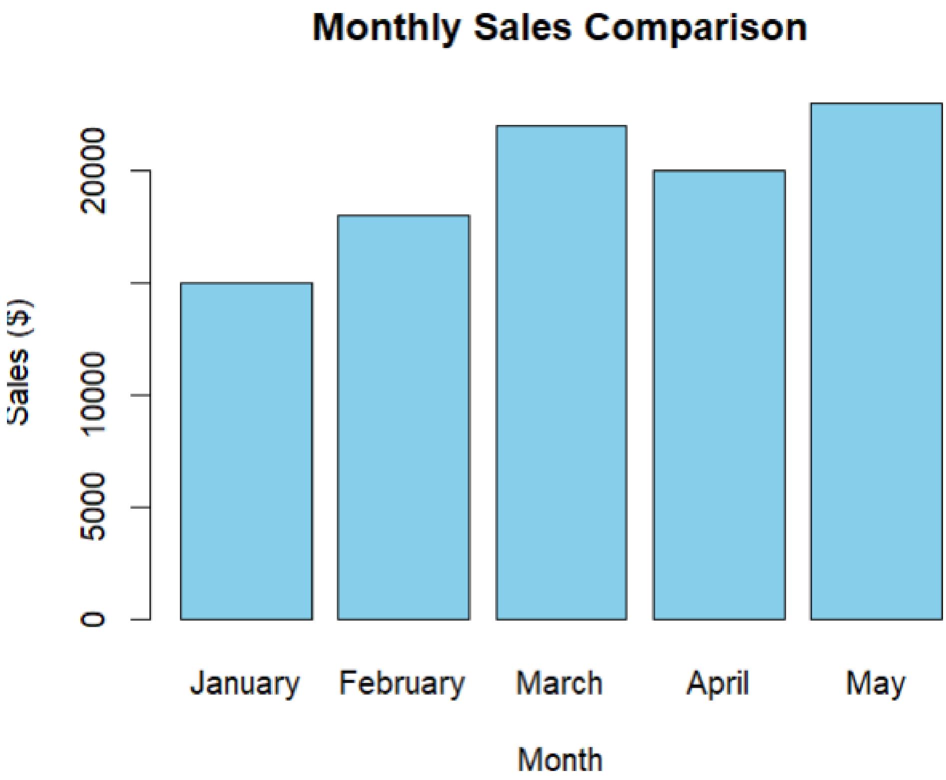


b) Generate a Bar Chart for Top-Selling Products

R code:

```
# Dataset
sales <- data.frame(
  Month = c("January", "February", "March", "April", "May"),
  Sales = c(15000, 18000, 22000, 20000, 23000)
)

# Bar Chart
barplot(sales$Sales,
        names.arg = sales$Month,
        col = "skyblue",
        xlab = "Month",
        ylab = "Sales ($)",
        main = "Monthly Sales Comparison")
```

output:



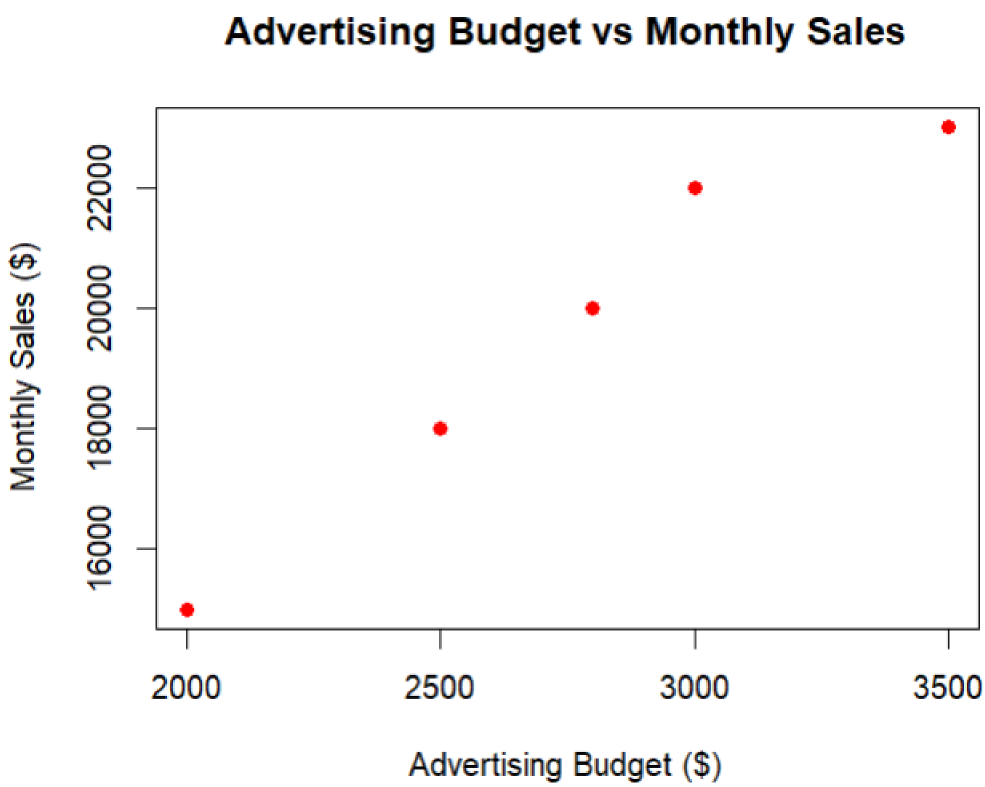
c) Scatter Plot Between Advertising Budget and Monthly Sales

R code:

```
# Dataset
sales <- data.frame(
  Month = c("January", "February", "March", "April", "May"),
  Sales = c(15000, 18000, 22000, 20000, 23000),
  Advertising = c(2000, 2500, 3000, 2800, 3500)
)

# Scatter Plot
plot(sales$Advertising,
     sales$Sales,
     pch = 19,
     col = "red",
     xlab = "Advertising Budget ($)",
```

```r
    ylab = "Monthly Sales ($)",
    main = "Advertising Budget vs Monthly Sales")

# Regression Line
abline(lm(Sales ~ Advertising, data = sales),
       col = "blue",
       lwd = 2)
```

output:



d) Interactive Dashboard (Using Shiny)

R code:

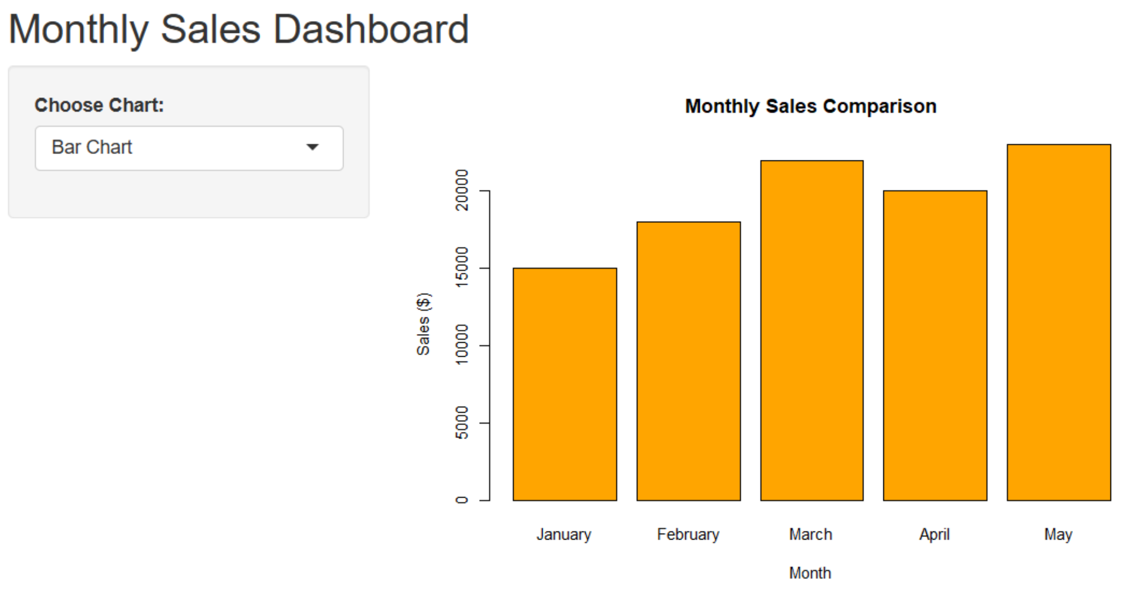
```r
library(shiny)

sales <- data.frame(
  Month = c("January", "February", "March", "April", "May"),
  Sales = c(15000, 18000, 22000, 20000, 23000)
```

```
)

ui <- fluidPage(

   titlePanel("Monthly Sales Dashboard"),

   sidebarLayout(

      sidebarPanel(

         selectInput("chart",
                  "Choose Chart:",
                  choices = c("Line Chart",
                           "Bar Chart"))

      ),

      mainPanel(

         plotOutput("salesPlot")

      )

   )

)

server <- function(input, output){
```

```
}

output$salesPlot <- renderPlot({

  if(input$chart == "Line Chart"){

    plot(sales$Sales,
         type = "o",
         xaxt = "n",
         xlab = "Month",
         ylab = "Sales ($)",
         main = "Monthly Sales Trend",
         col = "blue",
         lwd = 2)

    axis(1,
         at = 1:nrow(sales),
         labels = sales$Month)

  }else{

    barplot(sales$Sales,
            names.arg = sales$Month,
            col = "orange",
            xlab = "Month",
            ylab = "Sales ($)",
            main = "Monthly Sales Comparison")

  }

})
```

```
}

shinyApp(ui = ui, server = server)
```

output:



Set 2: Customer Feedback Analysis

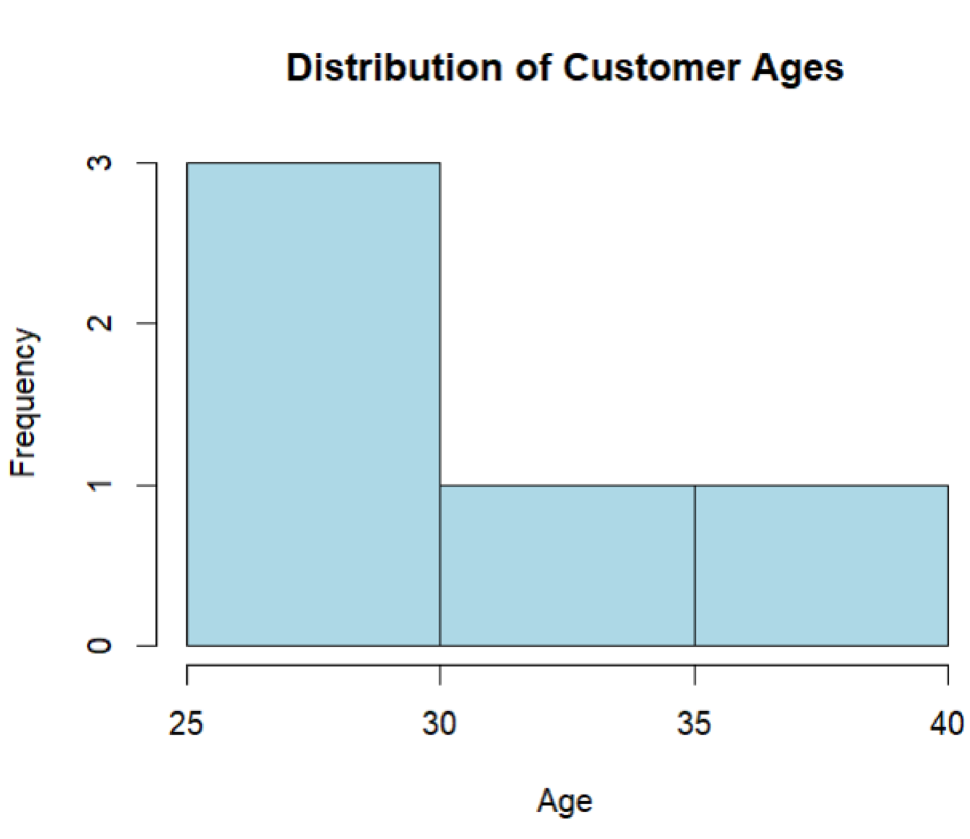
a) Create a Histogram for Customer Ages

R code:

```
# Dataset
customer <- data.frame(
  CustomerID = c(1, 2, 3, 4, 5),
  Age = c(25, 30, 35, 28, 40),
  Satisfaction = c(4, 5, 3, 4, 5)
```

```
)

# Histogram
hist(customer$Age,
     main = "Distribution of Customer Ages",
     xlab = "Age",
     ylab = "Frequency",
     col = "lightblue",
     border = "black",
     breaks = 5)
```

output:



b) Generate a Pie Chart for Satisfaction Scores

R code:

```
customer <- data.frame(
  CustomerID = c(1, 2, 3, 4, 5),
```

```
  Age = c(25, 30, 35, 28, 40),
  Satisfaction = c(4, 5, 3, 4, 5)
)

# Count satisfaction scores
score_count <- table(customer$Satisfaction)

# Pie Chart
pie(score_count,
    labels = paste(names(score_count), "Stars"),
    col = rainbow(length(score_count)),
    main = "Customer Satisfaction Scores")
```

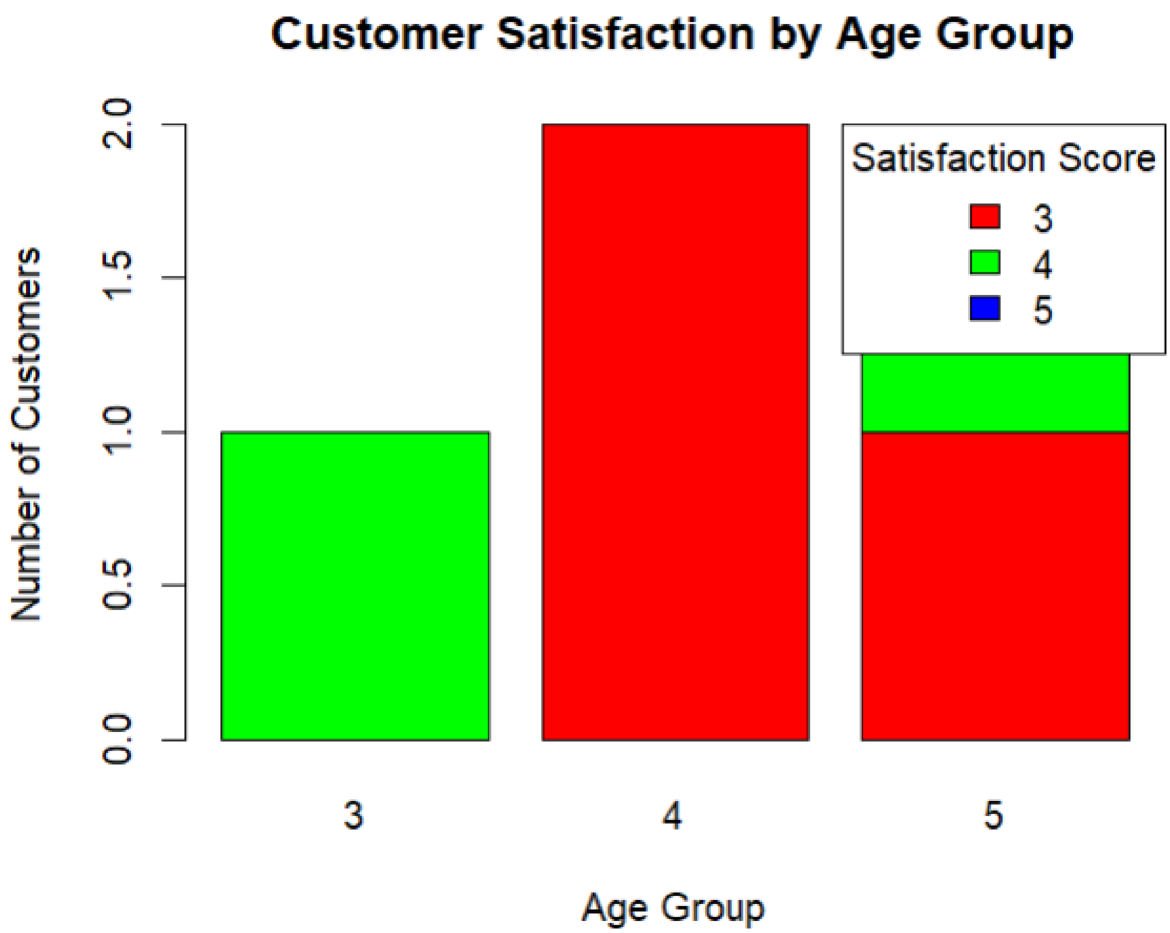
output:

c) Create a Stacked Bar Chart by Age Group

R code:

```
# Dataset
customer <- data.frame(
  CustomerID = c(1, 2, 3, 4, 5),
  Age = c(25, 30, 35, 28, 40),
  Satisfaction = c(4, 5, 3, 4, 5)
)

# Create Age Groups
customer$AgeGroup <- cut(customer$Age,
                         breaks = c(20, 30, 40, 50),
                         labels = c("21-30", "31-40", "41-50"))

# Frequency Table
table_data <- table(customer$AgeGroup, customer$Satisfaction)

# Stacked Bar Chart
barplot(table_data,
        beside = FALSE,
        col = rainbow(ncol(table_data)),
        main = "Customer Satisfaction by Age Group",
        xlab = "Age Group",
        ylab = "Number of Customers")

legend("topright",
       legend = colnames(table_data),
       fill = rainbow(ncol(table_data)),
       title = "Satisfaction Score")
```

output:



d) Create a Word Cloud from Customer Feedback

R code:

```
library(tm)
library(wordcloud)
library(RColorBrewer)

# Sample Feedback
feedback <- c(
  "Excellent service and friendly staff",
  "Very satisfied with the product quality",
  "Good experience but delivery was slow",
  "Excellent customer support",
```

```
  "Fast delivery and excellent quality"
)

# Create Corpus
corpus <- Corpus(VectorSource(feedback))

# Clean Text
corpus <- tm_map(corpus, content_transformer(tolower))
corpus <- tm_map(corpus, removePunctuation)
corpus <- tm_map(corpus, removeNumbers)
corpus <- tm_map(corpus, removeWords, stopwords("english"))

# Create Word Cloud
wordcloud(
  words = unlist(sapply(corpus, as.character)),
  max.words = 50,
  random.order = FALSE,
  colors = brewer.pal(8, "Dark2")
)
```

Output:

Set 3: Employee Performance Evaluation

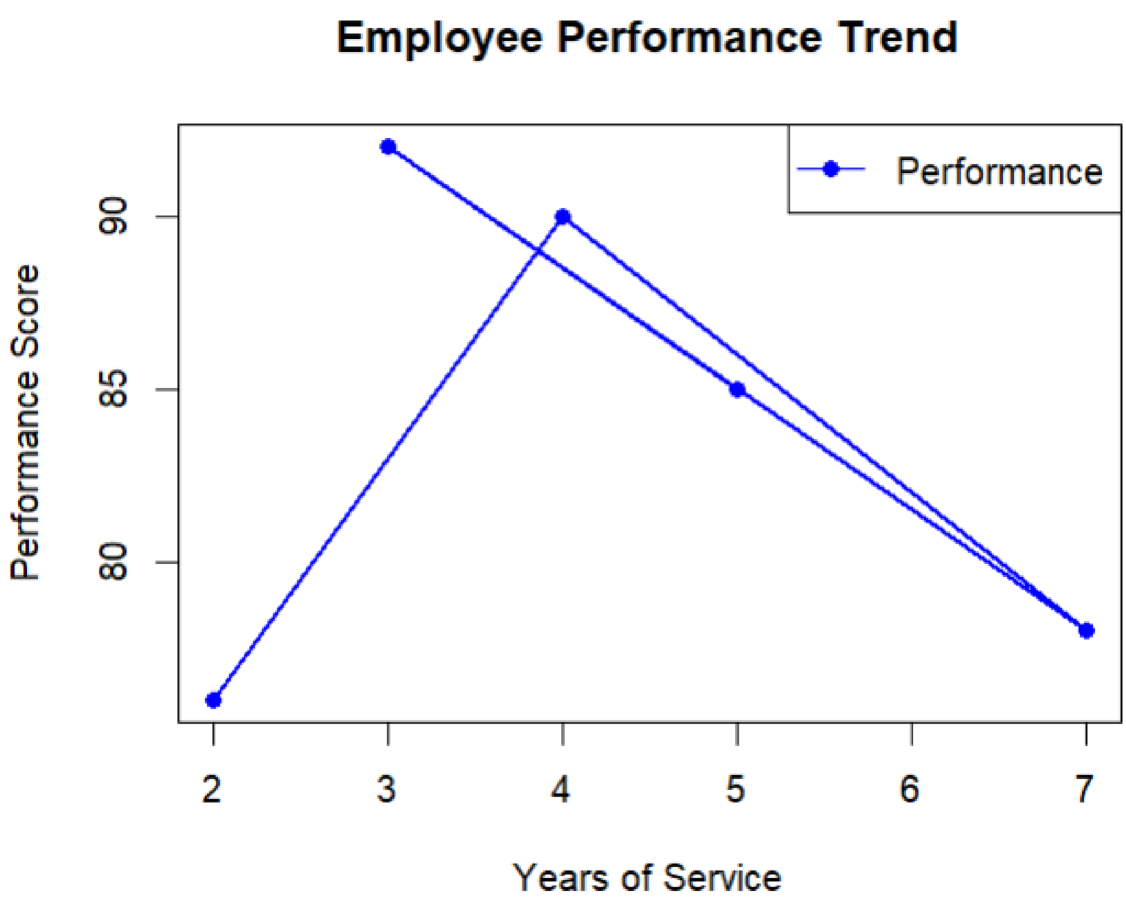
a) Create a Line Chart for Employee Performance Trend

R code:

```
# Dataset
employee <- data.frame(
  EmployeeID = c(1, 2, 3, 4, 5),
  Department = c("Sales", "HR", "Marketing", "Sales", "HR"),
  YearsService = c(5, 3, 7, 4, 2),
  Performance = c(85, 92, 78, 90, 76)
)

# Line Chart
plot(employee$YearsService,
     employee$Performance,
```

```
    type = "o",
    col = "blue",
    pch = 16,
    lwd = 2,
    xlab = "Years of Service",
    ylab = "Performance Score",
    main = "Employee Performance Trend")

legend("topright",
     legend = "Performance",
     col = "blue",
     lty = 1,
     pch = 16)
```

output:



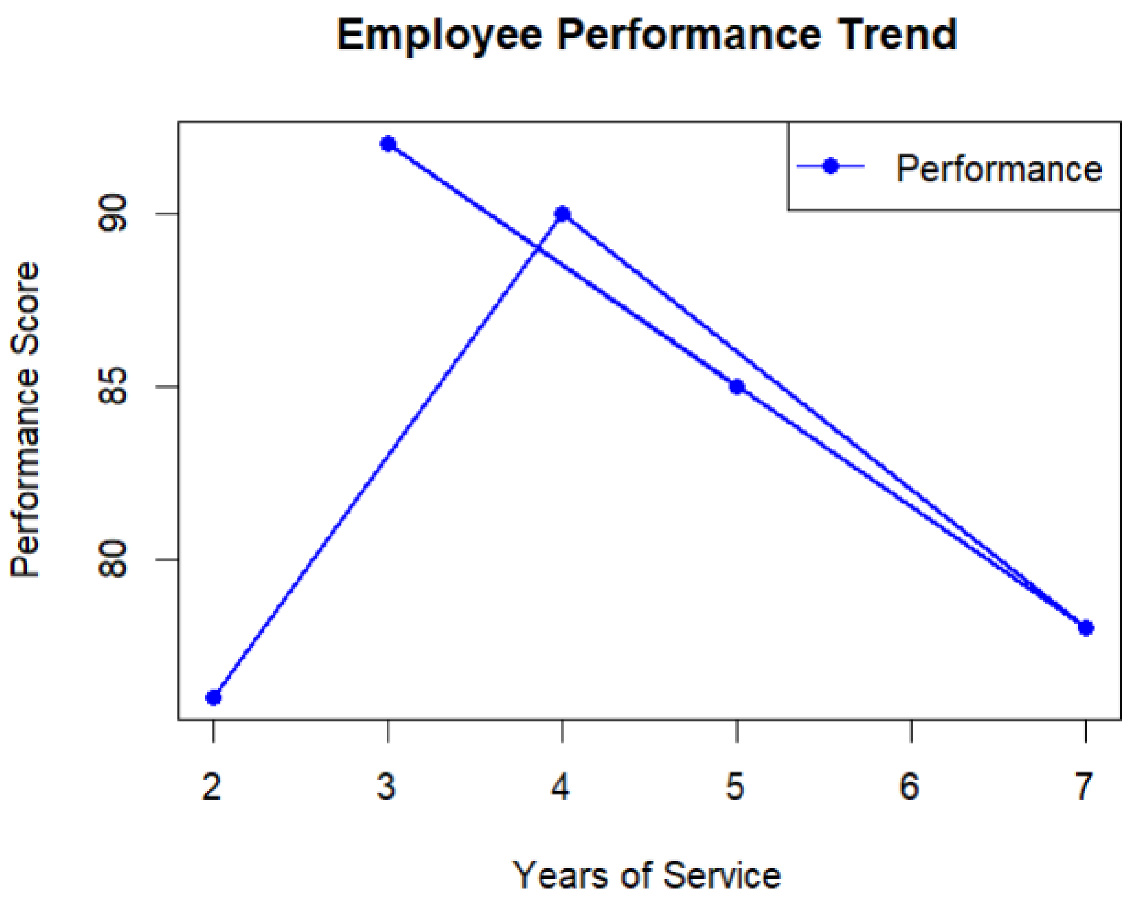
b) Generate a Bar Chart Showing Employees Across Departments

R code:

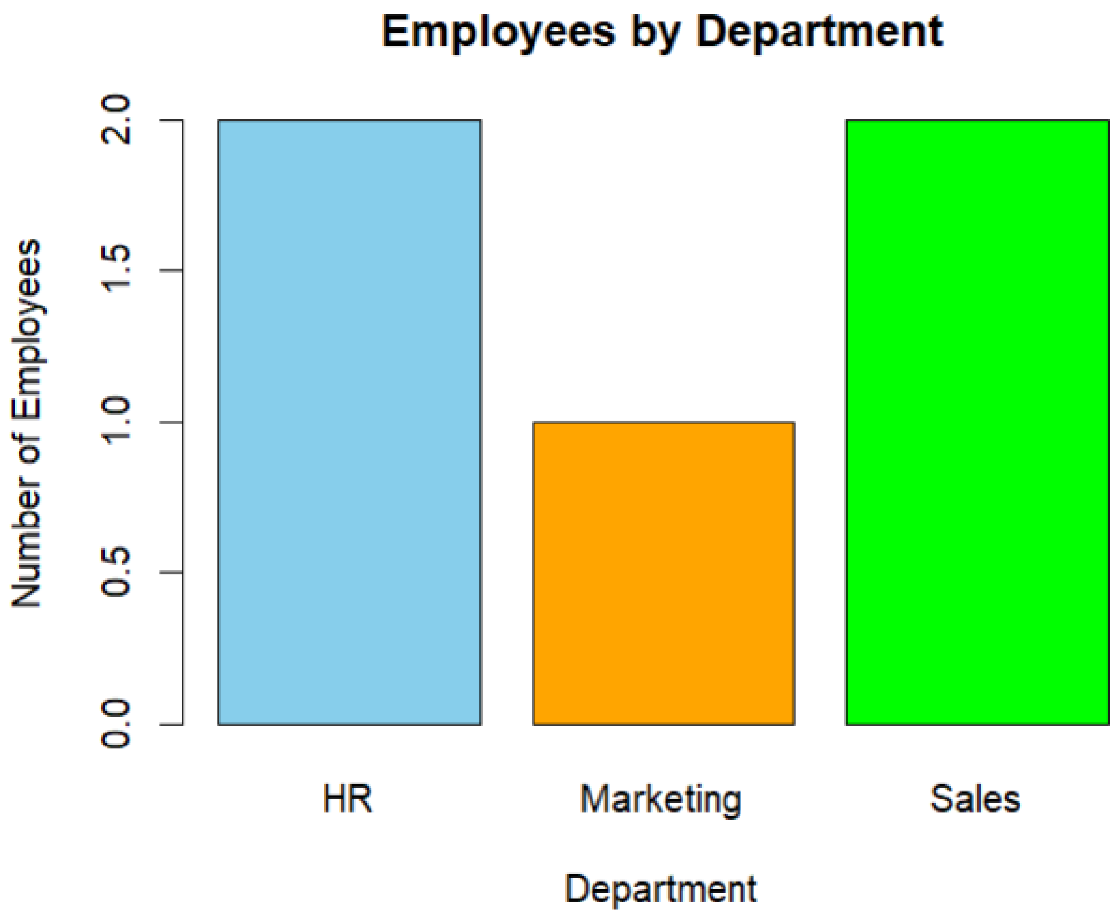
R code:

```
# Dataset
employee <- data.frame(
  EmployeeID = c(1, 2, 3, 4, 5),
  Department = c("Sales", "HR", "Marketing", "Sales", "HR"),
  YearsService = c(5, 3, 7, 4, 2),
  Performance = c(85, 92, 78, 90, 76)
)

# Line Chart
plot(employee$YearsService,
     employee$Performance,
     type = "o",
     col = "blue",
     pch = 16,
     lwd = 2,
     xlab = "Years of Service",
     ylab = "Performance Score",
     main = "Employee Performance Trend")

legend("topright",
       legend = "Performance",
       col = "blue",
       lty = 1,
       pch = 16)
```

output:



Output:



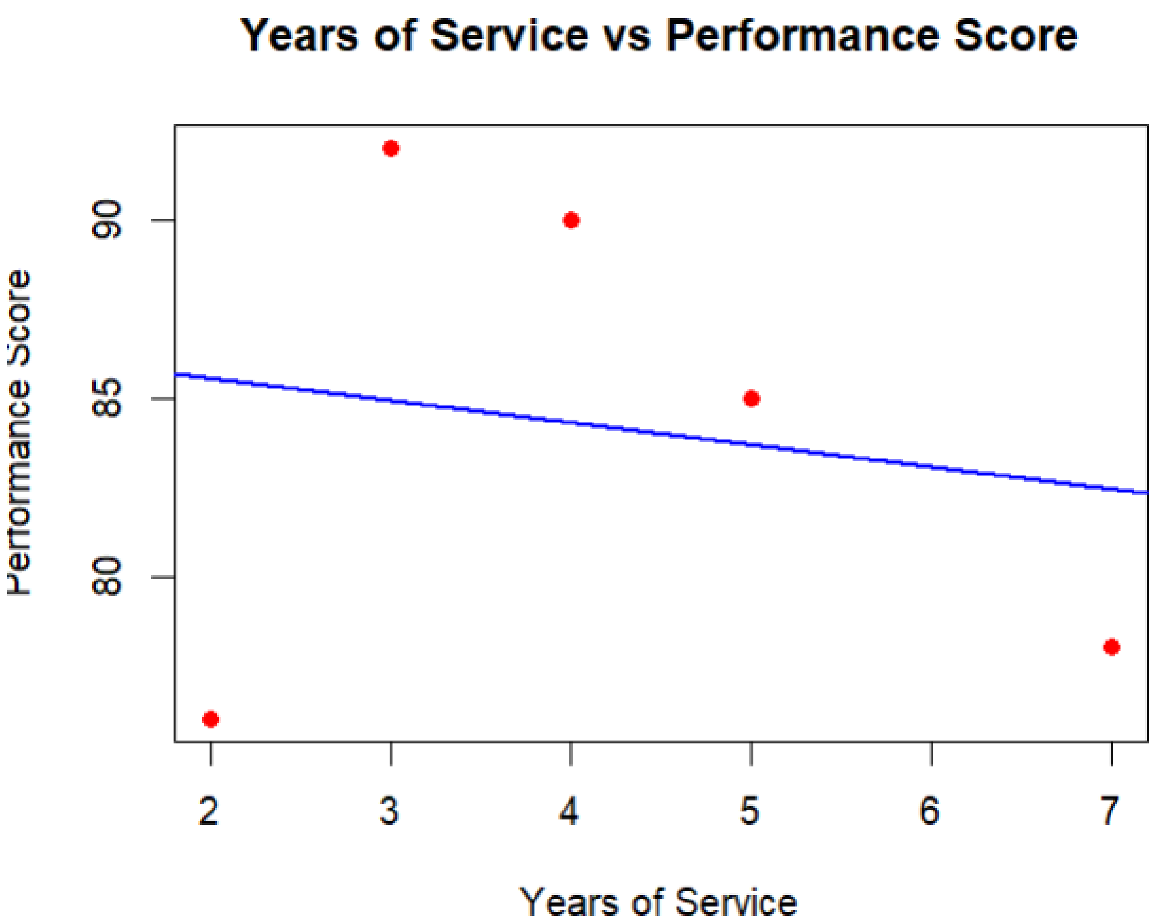
c) Create a Scatter Plot for Years of Service vs Performance Score

R code:

```
# Dataset
employee <- data.frame(
  EmployeeID = c(1, 2, 3, 4, 5),
  Department = c("Sales", "HR", "Marketing", "Sales", "HR"),
  YearsService = c(5, 3, 7, 4, 2),
  Performance = c(85, 92, 78, 90, 76)
)

# Scatter Plot
plot(employee$YearsService,
     employee$Performance,
```

```r
    pch = 19,
    col = "red",
    xlab = "Years of Service",
    ylab = "Performance Score",
    main = "Years of Service vs Performance Score")

# Regression Line
abline(lm(Performance ~ YearsService, data = employee),
       col = "blue",
       lwd = 2)
```

output:



d) Interactive Dashboard (Using Shiny)

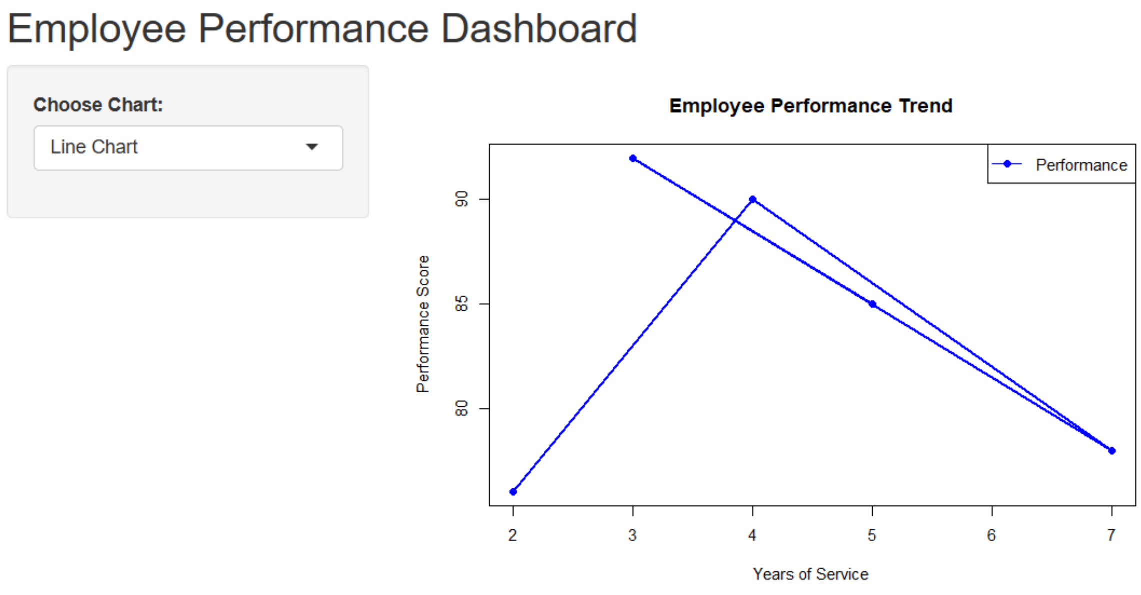
R code:

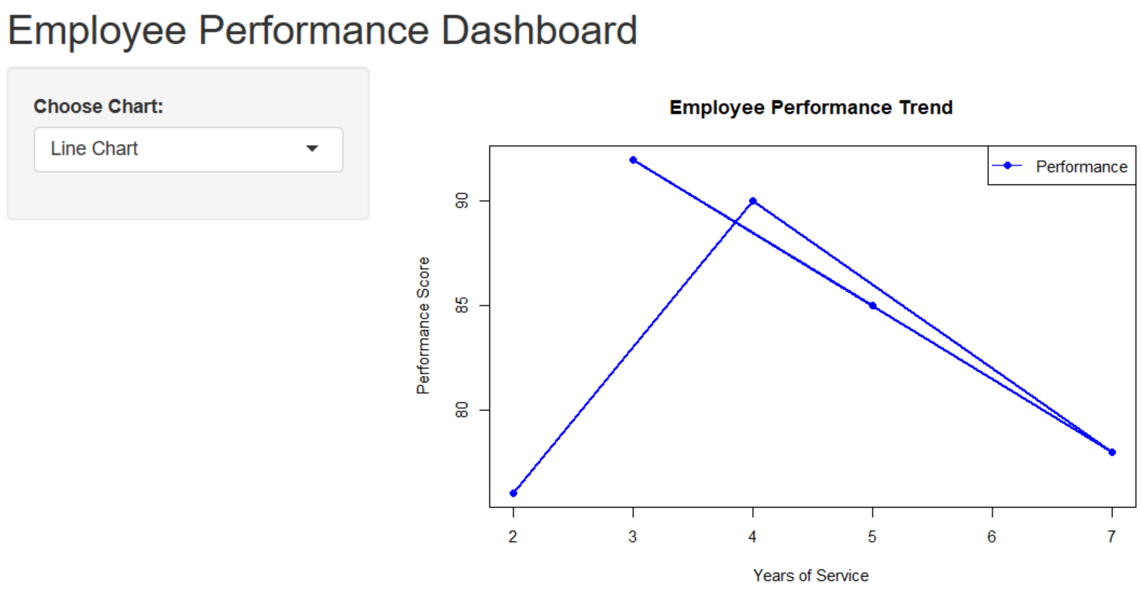
```r
library(shiny)

employee <- data.frame(
  EmployeeID = c(1, 2, 3, 4, 5),
  Department = c("Sales", "HR", "Marketing", "Sales", "HR"),
  YearsService = c(5, 3, 7, 4, 2),
  Performance = c(85, 92, 78, 90, 76)
)

ui <- fluidPage(

    titlePanel("Employee Performance Dashboard"),

    sidebarLayout(

        sidebarPanel(

            selectInput("chart",
                        "Choose Chart:",
                        choices = c("Line Chart",
                                    "Bar Chart"))

        ),

        mainPanel(

            plotOutput("employeePlot")
```

```
        )
    )
)

server <- function(input, output){

    output$employeePlot <- renderPlot({

        if(input$chart == "Line Chart"){

            plot(employee$YearsService,
                 employee$Performance,
                 type = "o",
                 col = "blue",
                 pch = 16,
                 lwd = 2,
                 xlab = "Years of Service",
                 ylab = "Performance Score",
                 main = "Employee Performance Trend")

            legend("topright",
                   legend = "Performance",
                   col = "blue",
                   lty = 1,
                   pch = 16)

        }else{
```

```
            dept_count <- table(employee$Department)

            barplot(dept_count,
                    col = c("skyblue", "orange", "green"),
                    xlab = "Department",
                    ylab = "Number of Employees",
                    main = "Employees by Department")

        }

    })

}

shinyApp(ui = ui, server = server)
```

output:


Employee Performance Dashboard
Choose Chart:
Line Chart
Employee Performance Trend
Performance
Performance Score
Years of Service
80
85
90
2
3
4
5
6
7



Set 4: Product Inventory Management

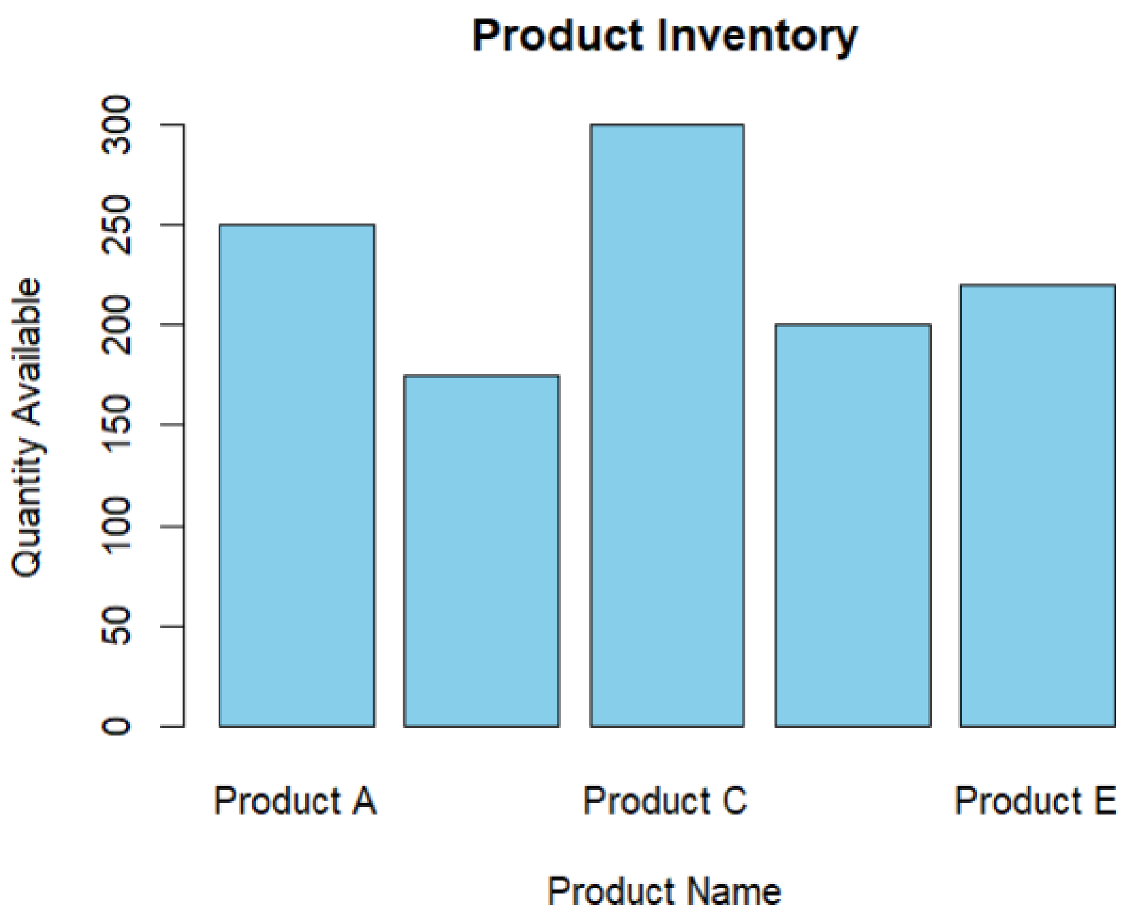
a) Create a Bar Chart for Product Inventory

R code: # Dataset

```
inventory <- data.frame(
  ProductID = c(1, 2, 3, 4, 5),
  ProductName = c("Product A", "Product B", "Product C", "Product D", "Product E"),
  Quantity = c(250, 175, 300, 200, 220)
)

# Bar Chart
barplot(inventory$Quantity,
        names.arg = inventory$ProductName,
        col = "skyblue",
```

```
    xlab = "Product Name",
    ylab = "Quantity Available",
    main = "Product Inventory")
```

output:



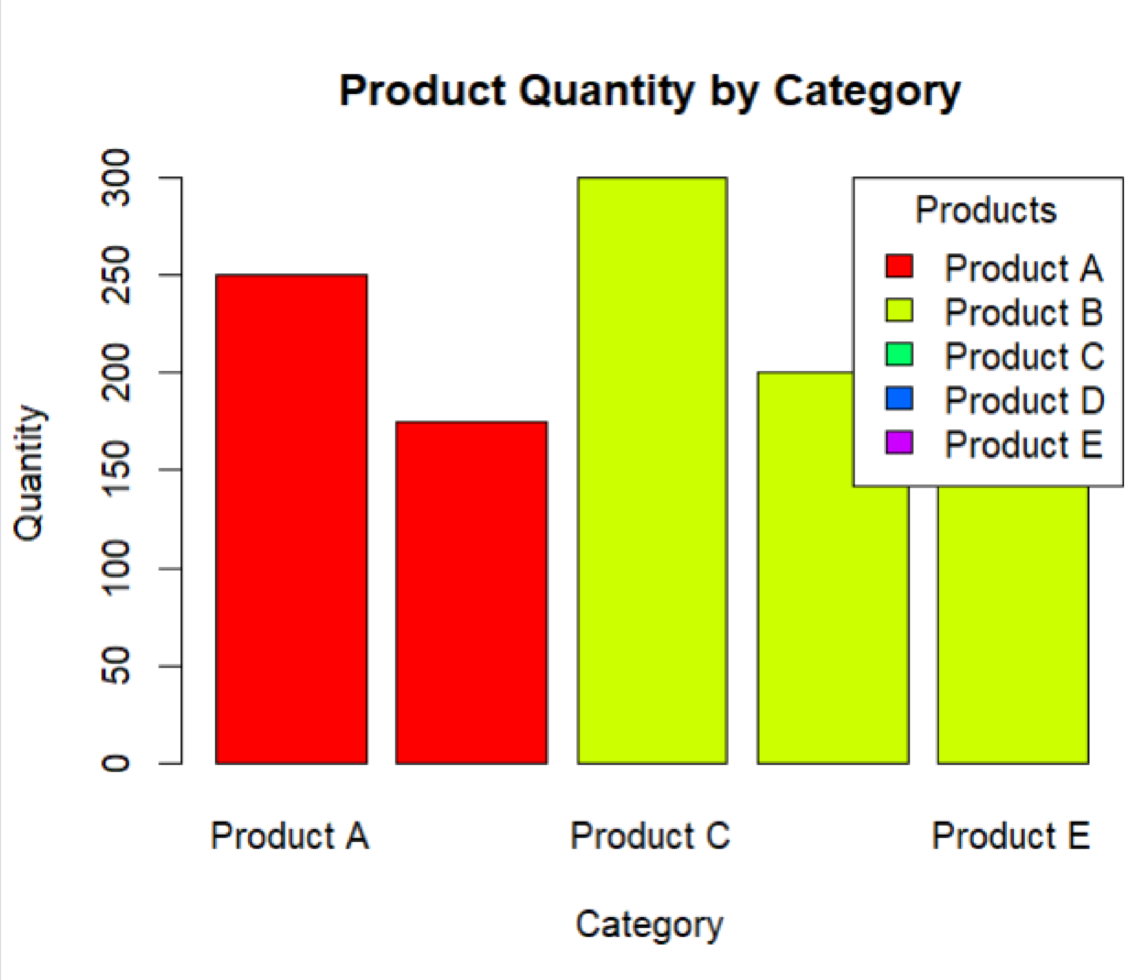
b) Generate a Stacked Bar Chart for Product Categories

R code:

```
# Dataset
inventory <- data.frame(
  ProductName = c("Product A", "Product B", "Product C", "Product D", "Product E"),
  Quantity = c(250, 175, 300, 200, 220),
  Category = c("Electronics", "Electronics", "Furniture", "Furniture", "Furniture")
)
```

```
# Create Table
table_data <- xtabs(Quantity ~ Category + ProductName, data = inventory)

# Stacked Bar Chart
barplot(table_data,
        beside = FALSE,
        col = rainbow(ncol(table_data)),
        main = "Product Quantity by Category",
        xlab = "Category",
        ylab = "Quantity")

legend("topright",
       legend = colnames(table_data),
       fill = rainbow(ncol(table_data)),
       title = "Products")
```

output:



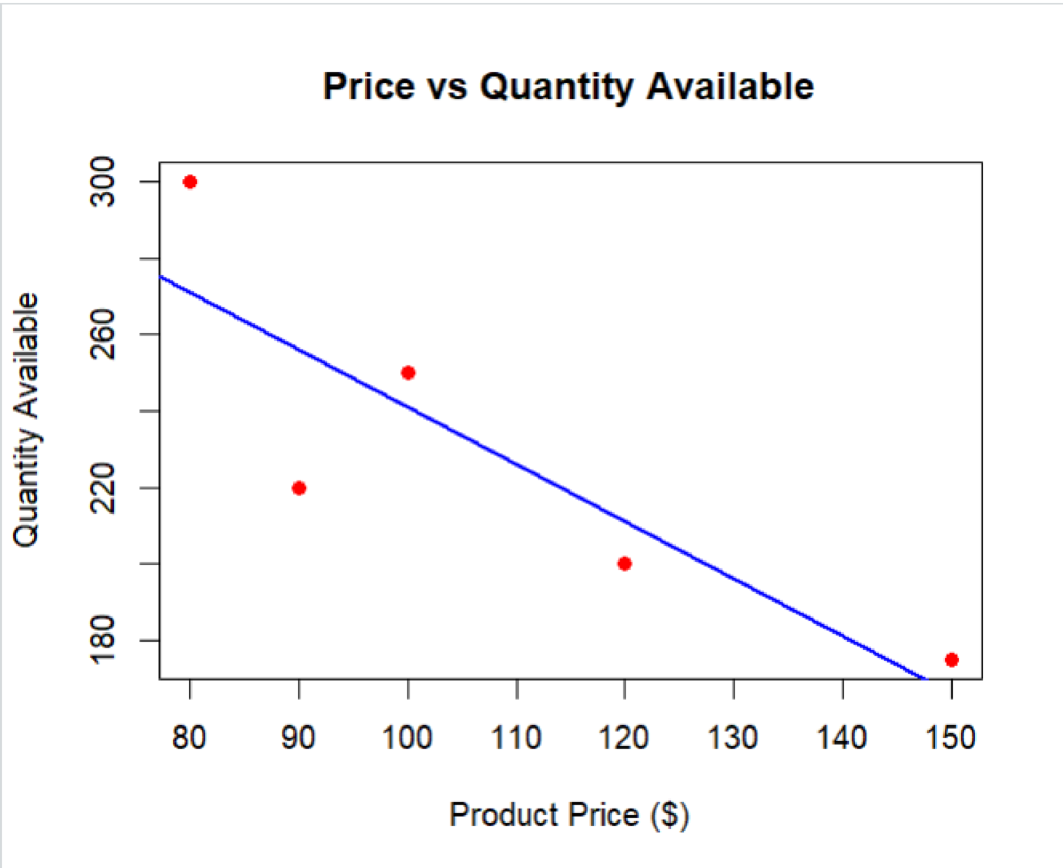
c) Create a Scatter Plot for Product Price vs Quantity

R code:

```
# Dataset
inventory <- data.frame(
  ProductName = c("Product A", "Product B", "Product C", "Product D", "Product E"),
  Quantity = c(250, 175, 300, 200, 220),
  Price = c(100, 150, 80, 120, 90)
)

# Scatter Plot
plot(inventory$Price,
     inventory$Quantity,
     pch = 19,
```

```
    col = "red",
    xlab = "Product Price ($)",
    ylab = "Quantity Available",
    main = "Price vs Quantity Available")

# Regression Line
abline(lm(Quantity ~ Price, data = inventory),
       col = "blue",
       lwd = 2)
```

output:



d) Interactive Dashboard (Using Shiny)

R code:

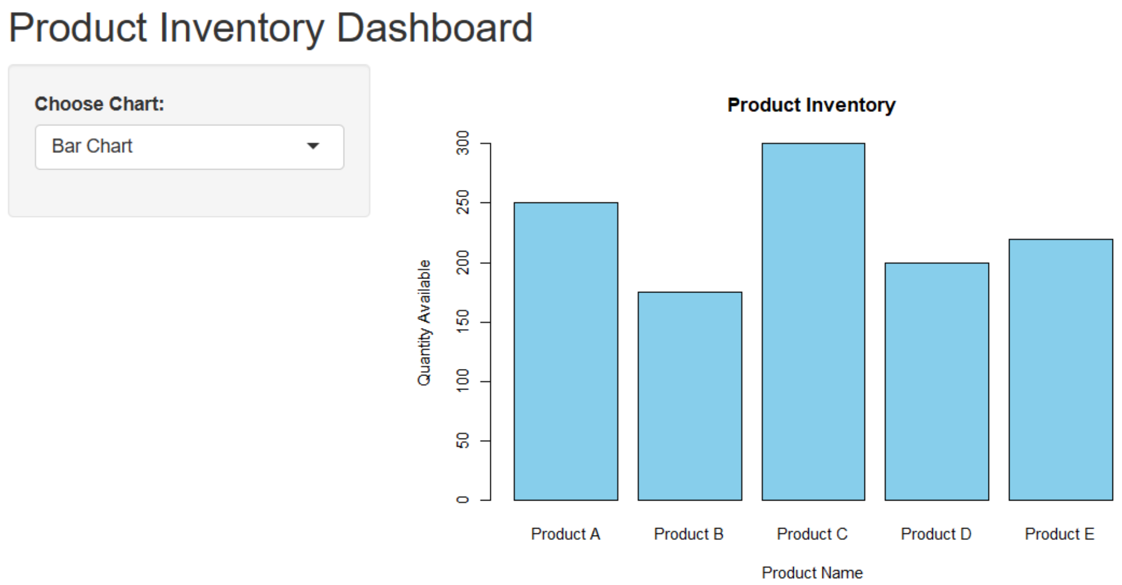
```
library(shiny)


inventory <- data.frame(
```

```r
  ProductID = c(1, 2, 3, 4, 5),
  ProductName = c("Product A", "Product B", "Product C", "Product D", "Product E"),
  Quantity = c(250, 175, 300, 200, 220),
  Category = c("Electronics", "Electronics", "Furniture", "Furniture", "Furniture")
)

ui <- fluidPage(

  titlePanel("Product Inventory Dashboard"),

  sidebarLayout(

    sidebarPanel(

      selectInput("chart",
            "Choose Chart:",
            choices = c("Bar Chart",
                  "Stacked Bar Chart"))

    ),

    mainPanel(

      plotOutput("inventoryPlot")

    )

  )
```

```
)

server <- function(input, output){

    output$inventoryPlot <- renderPlot({

        if(input$chart == "Bar Chart"){

            barplot(inventory$Quantity,
                    names.arg = inventory$ProductName,
                    col = "skyblue",
                    xlab = "Product Name",
                    ylab = "Quantity Available",
                    main = "Product Inventory")

        }else{

            table_data <- xtabs(Quantity ~ Category + ProductName,
                                data = inventory)

            barplot(table_data,
                    beside = FALSE,
                    col = rainbow(ncol(table_data)),
                    main = "Product Quantity by Category",
                    xlab = "Category",
                    ylab = "Quantity")

            legend("topright",
                   legend = colnames(table_data),
```

```
            fill = rainbow(ncol(table_data)),
            title = "Products")

      }

    })

}

shinyApp(ui = ui, server = server)
```

output:



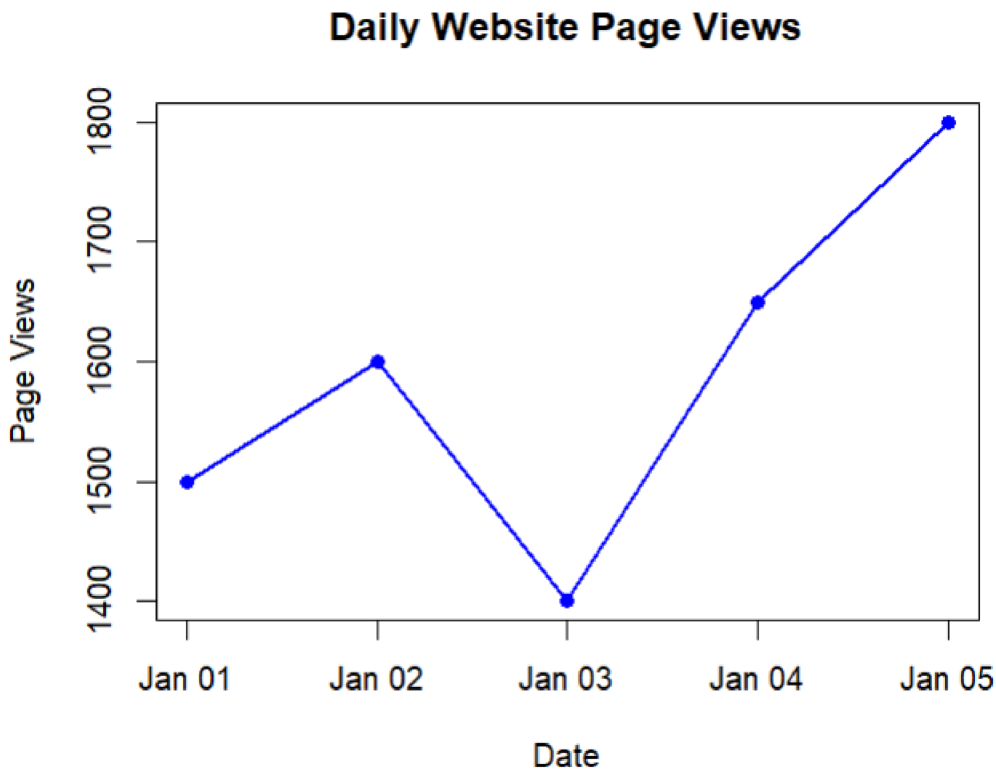
Set 5: Website Analytics

a) Create a Line Chart for Daily Page Views

R code:

```r
# Dataset
traffic <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03",
                 "2023-01-04", "2023-01-05")),
  PageViews = c(1500, 1600, 1400, 1650, 1800),
  CTR = c(2.3, 2.7, 2.0, 2.4, 2.6)
)

# Line Chart
plot(traffic$Date,
     traffic$PageViews,
     type = "o",
     col = "blue",
     pch = 16,
     lwd = 2,
     xlab = "Date",
     ylab = "Page Views",
     main = "Daily Website Page Views")
```

output:



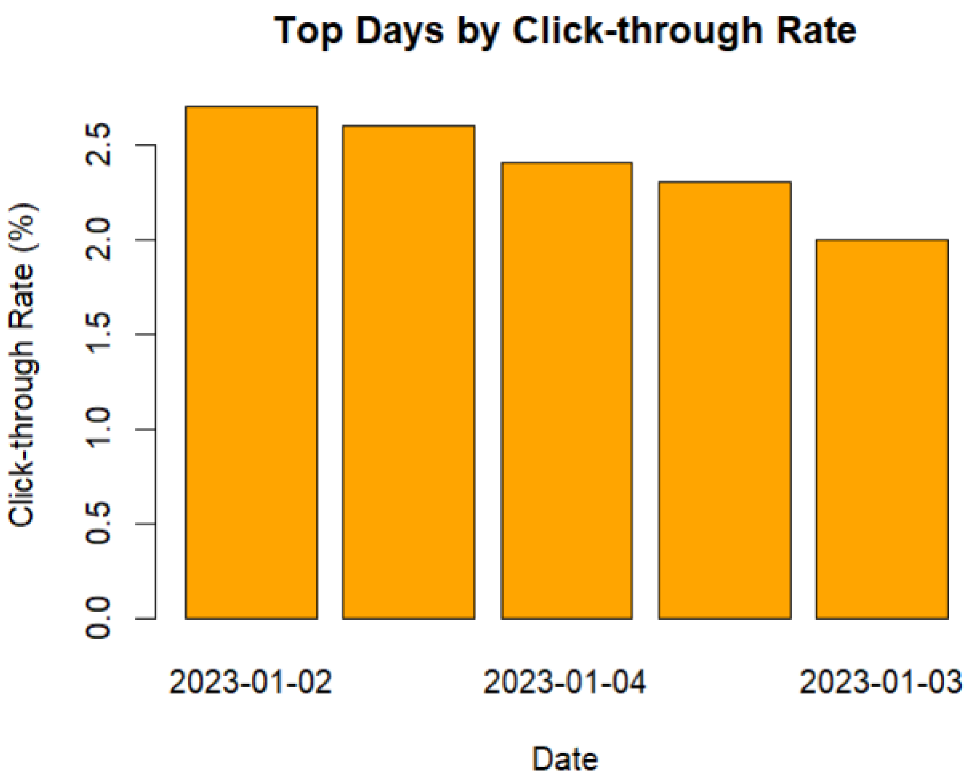
b) Generate a Bar Chart for Highest Click-through Rates

R code:

```
# Dataset
traffic <- data.frame(
  Date = c("2023-01-01", "2023-01-02", "2023-01-03",
           "2023-01-04", "2023-01-05"),
  PageViews = c(1500, 1600, 1400, 1650, 1800),
  CTR = c(2.3, 2.7, 2.0, 2.4, 2.6)
)

# Sort by CTR
traffic <- traffic[order(-traffic$CTR), ]

# Bar Chart
barplot(traffic$CTR,
```

```r
        names.arg = traffic$Date,
        col = "orange",
        xlab = "Date",
        ylab = "Click-through Rate (%)",
        main = "Top Days by Click-through Rate")
```

output:



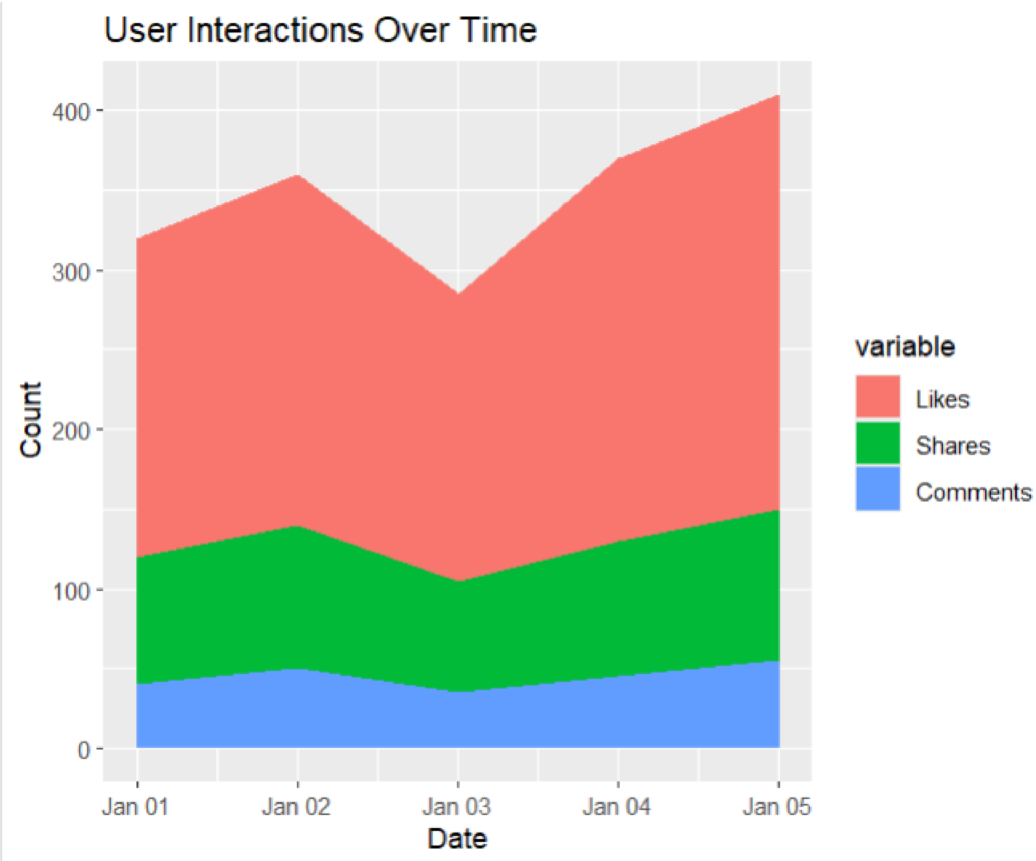
c) Create a Stacked Area Chart for User Interactions

R code:

```r
# Sample Dataset
traffic <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03",
                   "2023-01-04", "2023-01-05")),
  Likes = c(200, 220, 180, 240, 260),
  Shares = c(80, 90, 70, 85, 95),
  Comments = c(40, 50, 35, 45, 55)
```

```r
)

# Install package once
install.packages("ggplot2")
install.packages("reshape2")

library(ggplot2)
library(reshape2)

# Convert data to long format
traffic_long <- melt(traffic, id.vars = "Date")

# Stacked Area Chart
ggplot(traffic_long,
       aes(x = Date,
           y = value,
           fill = variable)) +
  geom_area() +
  labs(title = "User Interactions Over Time",
       x = "Date",
       y = "Count")
```

output:



d) Interactive Dashboard (Using Shiny)

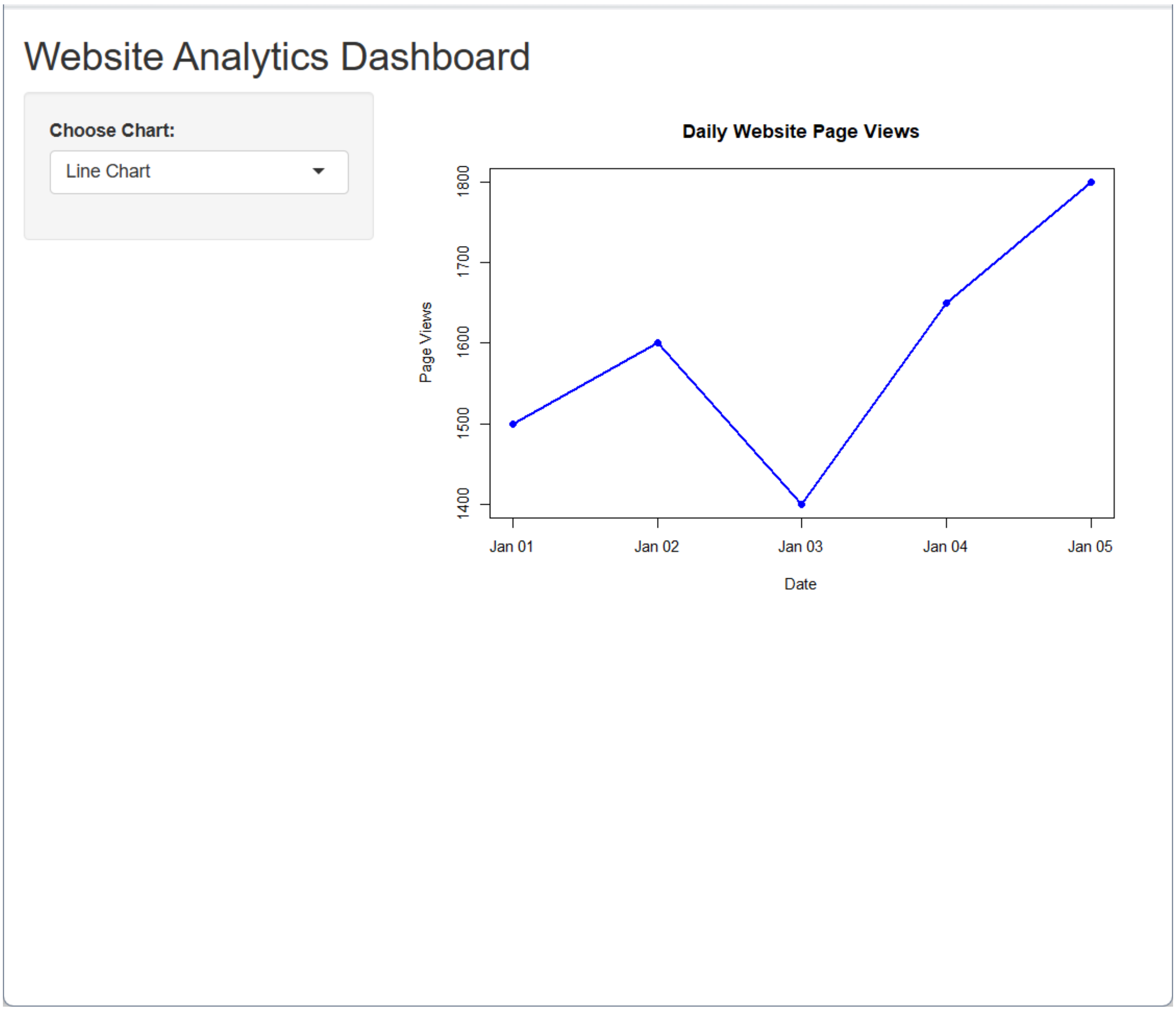
R code:

```
library(shiny)

traffic <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03",
                   "2023-01-04", "2023-01-05")),
  PageViews = c(1500, 1600, 1400, 1650, 1800),
  CTR = c(2.3, 2.7, 2.0, 2.4, 2.6)
)

ui <- fluidPage(

    titlePanel("Website Analytics Dashboard"),
```

```
  sidebarLayout(

    sidebarPanel(

      selectInput("chart",
            "Choose Chart:",
            choices = c("Line Chart",
                  "Bar Chart"))

    ),

    mainPanel(

      plotOutput("trafficPlot")

    )

  )

)

server <- function(input, output){

  output$trafficPlot <- renderPlot({

    if(input$chart == "Line Chart"){

      plot(traffic$Date,
        traffic$PageViews,
```

```
            type = "o",
            col = "blue",
            pch = 16,
            lwd = 2,
            xlab = "Date",
            ylab = "Page Views",
            main = "Daily Website Page Views")

      }else{

        temp <- traffic[order(-traffic$CTR), ]

        barplot(temp$CTR,
              names.arg = temp$Date,
              col = "orange",
              xlab = "Date",
              ylab = "Click-through Rate (%)",
              main = "Top Days by CTR")

      }

  })

}

shinyApp(ui = ui, server = server)
```

output:


Website Analytics Dashboard
Choose Chart:
Line Chart
Daily Website Page Views
Page Views
1400
1500
1600
1700
1800
Jan 01
Jan 02
Jan 03
Jan 04
Jan 05
Date